Practical No: 03

Title:

Enterprise architecture using **TOGAF** for Banking, Healthcare domain.

Problem Statement:

Design and Implement enterprise architecture using TOGAF for banking / healthcare domain.

Software Requirement: EDrawMax, Draw.io

Theory:

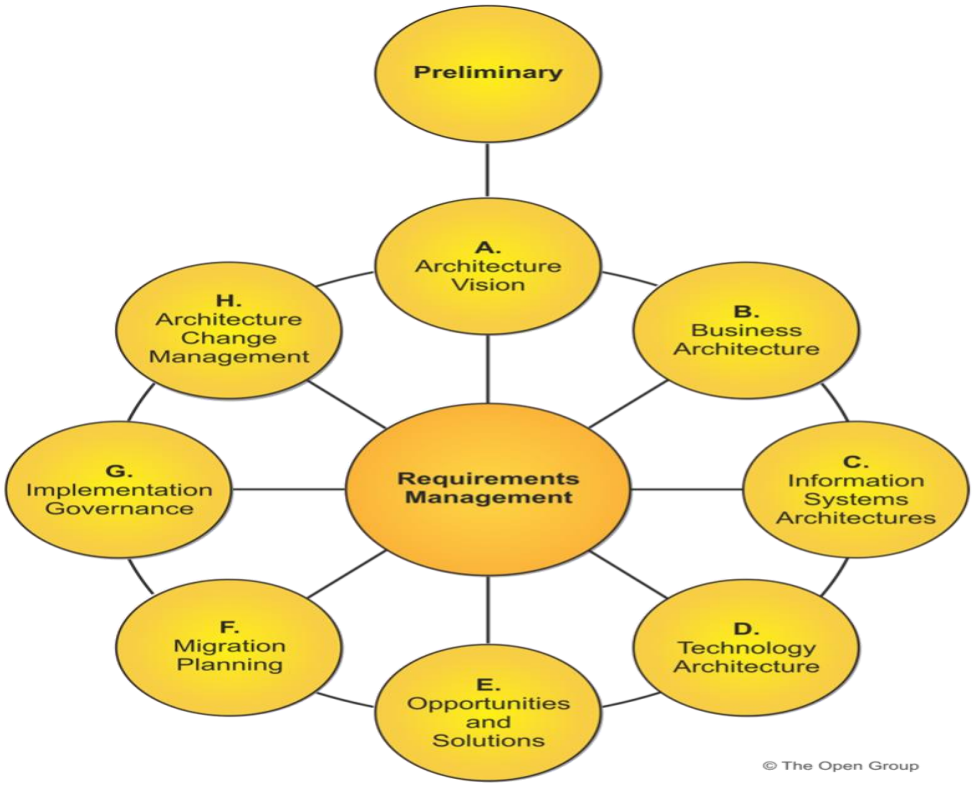
TOGAF (The Open Group Architecture Framework) is a globally recognized Enterprise Architecture (EA) framework developed by The Open Group. It provides a structured methodology, tools, best practices, and guidelines for designing, planning, implementing, and governing enterprise architecture. TOGAF helps organizations align business goals with IT strategies, improve decision-making, and manage change effectively.

Core Architectural Domains

- **Business Architecture:** Maps operational strategies, organizational structure, and key business processes.
- **Application Architecture:** Defines individual software applications and how they interact with each other.
- **Data Architecture:** Manages how a company stores, handles, and protects its data rules.
- **Technology Architecture:** Sets up the physical hardware, networks, and cloud infrastructure required

Objectives of TOGAF

- Align business strategy with IT strategy.
- Improve business efficiency and productivity.
- Standardize enterprise architecture practices.
- Reduce development costs and risks.
- Support digital transformation.
- Improve communication among stakeholders.
- Enable scalable and flexible enterprise systems.



Key Components of TOGAF

1. Architecture Development Method (ADM)

The **Architecture Development Method (ADM)** is the core of TOGAF. It provides a step-by-step, iterative process for developing and managing Enterprise Architecture.

2. Enterprise Continuum

A repository of reusable architecture assets, standards, templates, and best practices.

3. Architecture Content Framework

Defines the **deliverables, artifacts, and building blocks** required during architecture development.

4. Reference Models

TOGAF provides reusable reference models to simplify architecture design.

5. Architecture Capability Framework

Defines the governance structure, processes, roles, skills, and responsibilities needed to manage Enterprise Architecture successfully.

Phases of TOGAF ADM:

Preliminary Phase

The Preliminary Phase prepares the organization for Enterprise Architecture. It establishes the architecture capability, governance, principles, and scope before the architecture project begins.

Activities

- Define Enterprise Architecture scope.
- Identify stakeholders.
- Establish architecture governance.
- Create architecture principles.
- Select architecture tools and frameworks.
- Form the Architecture Team.

Phase A – Architecture Vision

To develop a high-level vision of the future architecture and gain stakeholder approval.

Activities

- Understand business goals.
- Identify stakeholders and their concerns.
- Define project scope.
- Prepare the business case.
- Develop the Architecture Vision.
- Obtain management approval.

Phase B – Business Architecture

To define the organization's business processes, organizational structure, capabilities, and business services.

Activities

- Analyze current (As-Is) business processes.
- Design future (To-Be) business processes.
- Identify business capabilities.
- Perform gap analysis.
- Define business functions and services.

Phase C – Information Systems Architecture

This phase consists of **Data Architecture** and **Application Architecture**.

(a) Data Architecture

To define how enterprise data will be stored, managed, shared, and secured.

Activities

- Identify data entities.
- Create data models.
- Define data ownership.
- Develop data governance policies.
- Design databases.

(b) Application Architecture

To define the software applications required and how they interact with each other.

Activities

- Identify applications.
- Define application interactions.
- Design system integration.
- Select software solutions.

Phase D – Technology Architecture

To define the hardware, software, networks, cloud services, and infrastructure required to support the applications.

Activities

- Select technology platforms.
- Design network architecture.
- Define cloud infrastructure.
- Plan security mechanisms.
- Select middleware and operating systems.

Phase E – Opportunities and Solutions

To identify the best implementation options and develop solution strategies.

Activities

- Identify implementation projects.
- Evaluate technology options.
- Prioritize solutions.
- Define Transition Architecture.

- Estimate costs and benefits.

Phase F – Migration Planning

To create a roadmap for moving from the current architecture to the target architecture.

Activities

- Prioritize projects.
- Estimate budget and timelines.
- Allocate resources.
- Develop migration schedule.
- Perform risk analysis.

Phase G – Implementation Governance

To ensure that implementation follows the approved architecture and standards.

Activities

- Monitor project implementation.
- Conduct architecture compliance reviews.
- Resolve architecture issues.
- Verify adherence to standards.
- Review project deliverables.

Phase H – Architecture Change Management

To ensure the Enterprise Architecture remains up to date with changing business and technology needs.

Activities

- Monitor business changes.
- Evaluate new technologies.
- Process changes requests.
- Update architecture documents.
- Improve architecture continuously.

Requirements Management (Continuous Phase)

Requirements Management supports every ADM phase by ensuring that business and technical requirements are continuously collected, analyzed, validated, and updated.

Activities

- Gather requirements from stakeholders.
- Analyze and prioritize requirements.
- Validate requirements.
- Track changes throughout the project.
- Communicate updated requirements to all teams.

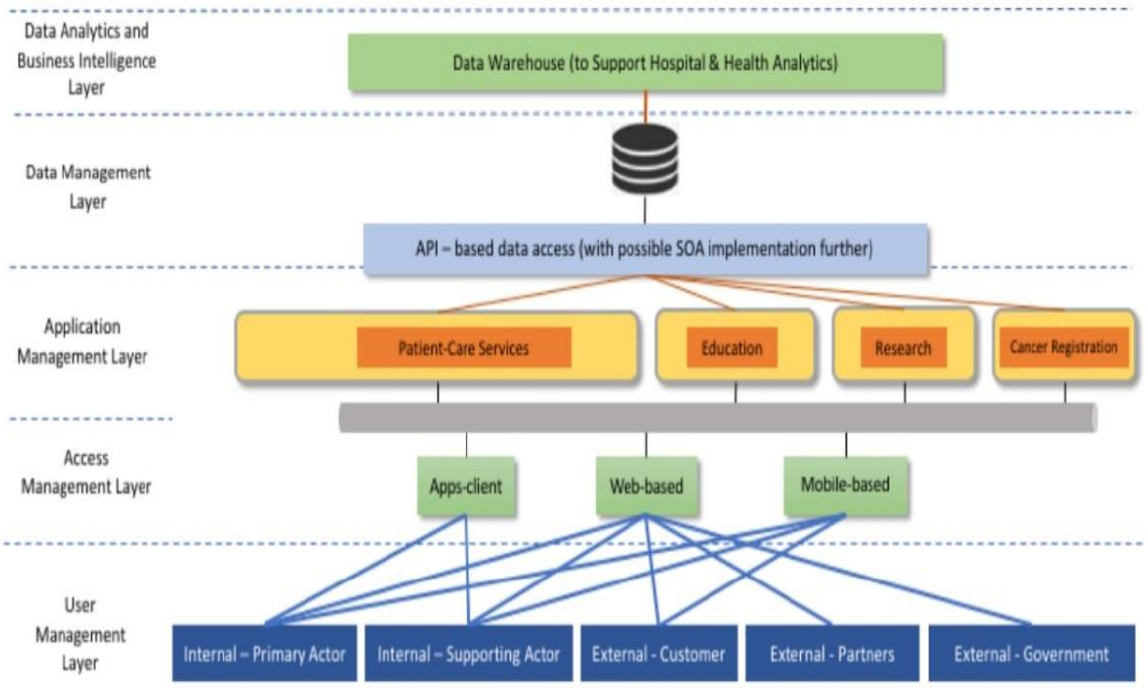


Fig: EA For Healthcare Domain Using TOGAF

Conclusion:

In this way, TOGAF enables enterprises to build a flexible, resilient, and future-ready healthcare architecture that supports long-term business growth and innovation.